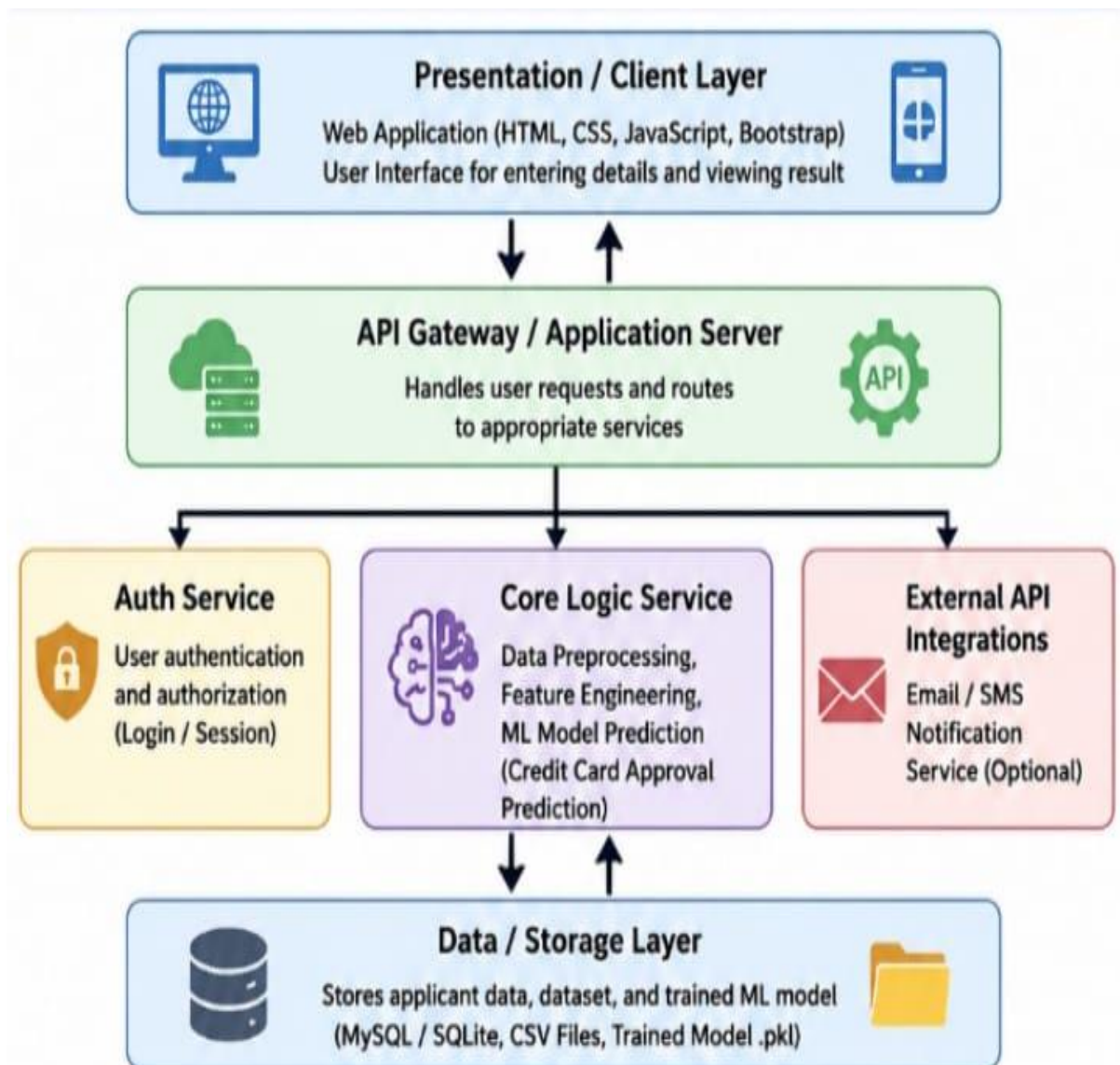


Solution Architecture

Date	10/07/2026
Name	Jaswanth Madapuram
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Solution Architecture Diagram:



Component Description Table:

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	User interface where applicants enter personal, financial, and employment details. Displays approval prediction and confidence score.	React.js, HTML, CSS, Bootstrap
API Gateway / Backend	Receives requests from the frontend, validates inputs, communicates with the ML model, and returns prediction results.	FastAPI / Flask, REST API
Authentication Service	Handles user login, authorization, and secure access to the application.	JWT, OAuth2 (optional), FastAPI Security
Core Logic Service	Cleans data, performs feature engineering, loads trained ML model, predicts credit card approval, and generates risk score.	Python, Pandas, NumPy, Scikit-learn, Joblib
External API Integrations	Sends email notifications or optionally retrieves applicant credit information from third-party services.	SMTP, Twilio, Credit Bureau API (Optional)
Database / Storage Layer	Stores applicant records, prediction history, trained model files, and datasets.	PostgreSQL/MySQL, CSV, Joblib (.pkl)